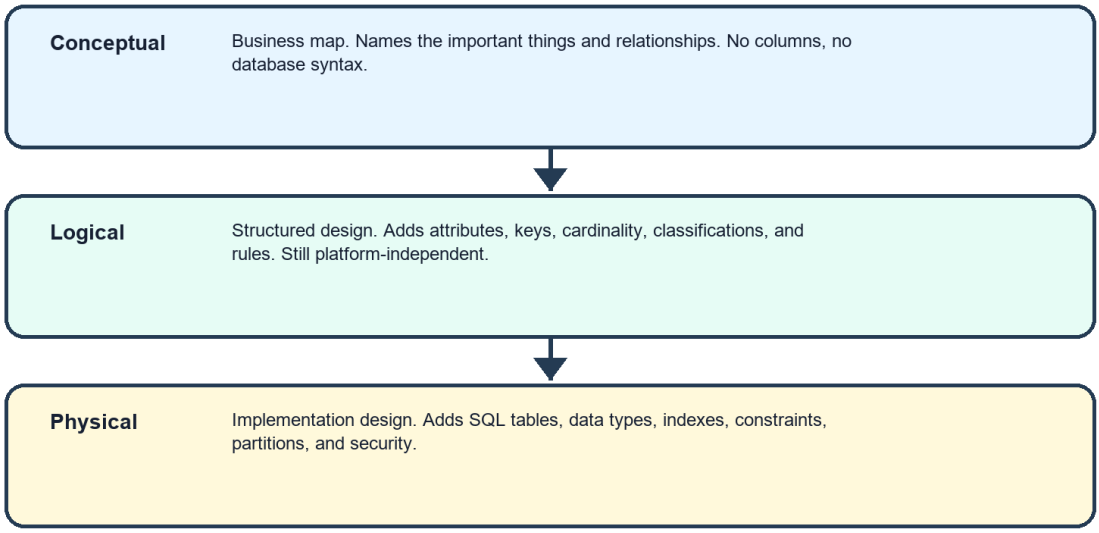
Traditional Data Modeling Teaching Guide

A focused, teaching-first guide to conceptual, logical, and physical data models.

Conceptual vs Logical vs Physical Models



The goal is simple: learn how data modelers move from business meaning to database implementation without losing clarity between the layers.

Sources used: Practical Data Modeling's traditional data model example and ScaleFree's Conceptual vs Logical vs Physical Data Models article.

1. What Data Modeling Is

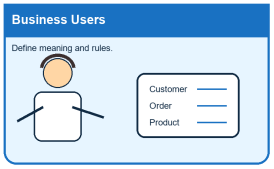
Data modeling is the practice of describing business information clearly enough that people and systems can use it consistently. It is not only a diagramming activity. It is a translation activity: business language becomes structure, and structure becomes implementation.

A good model answers three kinds of questions. First, what does the business care about? Second, how do those things relate? Third, how should the data be stored, protected, and queried?

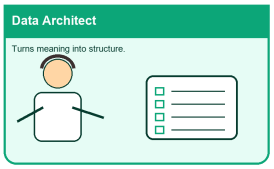
The Same Example Throughout

This guide uses a small commerce example: customers place orders, orders contain products, and payments settle orders. It is intentionally ordinary so the modeling ideas stay visible.

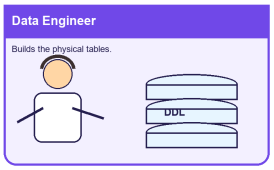
Who Cares About The Model



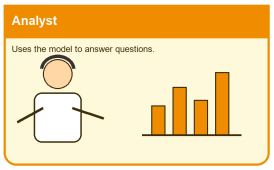
Business
Defines the meaning of words such as customer, order, and product.



Architect
Turns business meaning into entities, relationships, and rules.



Engineer
Implements tables, keys, constraints, indexes, and security.



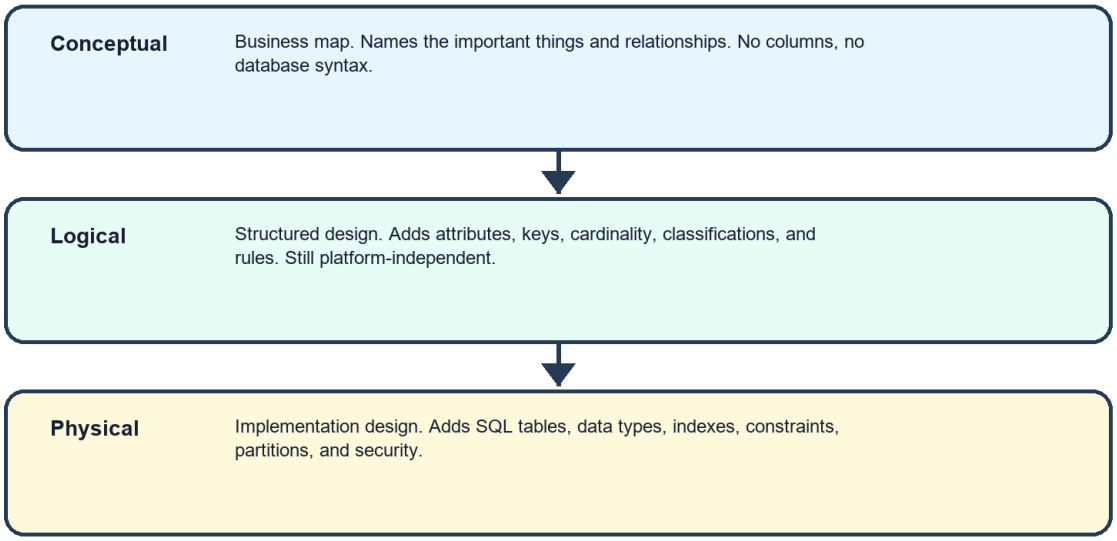
Analyst
Uses the model to produce trusted reports and analysis.

Quick check: In one sentence, explain why data modeling is more than drawing tables.

2. The Three Layers

ScaleFree uses a helpful building analogy: you do not start construction by randomly hammering wood together. You start with a sketch, then a blueprint, then the actual build. Data modeling works the same way.

Conceptual vs Logical vs Physical Models



Layer Comparison

Layer	Purpose	Focus	Audience
Conceptual	Shared business understanding	Entities and relationships	Business users, analysts, architects
Logical	Technology-independent structure	Attributes, keys, cardinality, rules, classifications	Architects, engineers, modelers
Physical	Actual database implementation	DDL, data types, constraints, indexes, storage, security	Engineers, DBAs, developers

The three layers are not competing approaches. They are steps in a process. Conceptual explains the world. Logical organizes it. Physical builds it.

Quick check: Which layer would you use when talking to non-technical stakeholders, and why?

3. Conceptual Model

The conceptual model is the high-level business map. It does not list every column. It does not care whether the future database is PostgreSQL, Snowflake, SQL Server, or something else. It captures the important business things and the relationships between them.

Commerce Example

Business sentence	Conceptual interpretation
A customer places an order.	Customer and Order are entities. Places is a relationship.
An order contains products.	Order and Product are related.
An order has a payment.	Payment is another business thing connected to Order.

Conceptual Modeling Vocabulary

Term	Plain meaning
Entity	A business thing worth tracking, such as Customer or Order.
Relationship	How entities are connected, such as Customer places Order.
Business rule	A rule that must remain true, such as one order must have a customer.
Semantic meaning	The agreed meaning of a word. Example: does customer mean registered user or paying buyer?

This is where you slow down and ask what words mean. Many data problems begin when two departments use the same word differently.

Quick check: Write two possible meanings of the word customer.

4. Logical Model

The logical model takes the conceptual model and makes it more precise. It adds attributes, identifiers, cardinality, and rules, but it still avoids platform-specific syntax.

Logical Entities

Entity	Attributes	Identifier
Customer	name, email, created date	customer_id
Order	order date, status, total amount	order_id
Product	sku, name, price, category	product_id
OrderItem	quantity, unit price	order_id + product_id
Payment	amount, method, payment date	payment_id

Why OrderItem Appears

One order can contain many products, and one product can appear in many orders. That is a many-to-many relationship. The logical model resolves it with OrderItem, a junction entity that connects Order and Product while storing the quantity.

Data Vault Note

In a Data Vault style model, the logical layer often classifies structures as hubs, links, and satellites. Hubs represent core business keys, links represent relationships, and satellites hold descriptive details and history. The idea is still the same: business meaning becomes structured design before it becomes physical tables.

Quick check: Why does quantity belong on OrderItem rather than Product?

5. Physical Model

The physical model is where the design becomes real database implementation. It includes table names, data types, primary keys, foreign keys, indexes, constraints, partitions, and security choices.

Logical To Physical Examples

Logical idea	Physical implementation example
Customer has email	customers.email VARCHAR(255) UNIQUE NOT NULL
Order belongs to Customer	orders.customer_id INTEGER REFERENCES customers(customer_id)
Quantity must be positive	CHECK (quantity > 0)
Order date is queried often	INDEX on orders.order_date
Sensitive data needs protection	Column masking, roles, restricted views, or separate secure table

Physical choices depend on the target platform. A Snowflake implementation, a PostgreSQL implementation, and an Oracle implementation may all represent the same logical model differently.

ScaleFree also points out that this is where platform-specific configuration, access patterns, and security become concrete. Logical says what the data means; physical says how it is stored and controlled.

Quick check: Give one example of a physical decision that does not belong in the conceptual model.

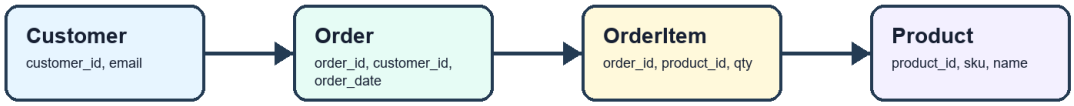
6. How A Source Table Evolves

Existing data often arrives as a flat extract. The modeler studies repeated values and mixed concepts, then separates them into entities with clear responsibilities.

How one source table becomes a model

Source extract: one useful but mixed table

order_id	customer_email	product_sku	product_name	qty	amount
1001	dana@example.com	SKU-RED-01	Red Shoes	2	119.98
1001	dana@example.com	SKU-LACE-02	Laces	1	4.99
1002	sam@example.com	SKU-BAG-07	Travel Bag	1	89.00



The model keeps the original facts, but puts each fact in the right home.

What Changes And What Stays

In the source table	In the model
customer_email repeats on every order line	Customer stores the email once.
product_name repeats for every order line	Product stores product facts once.
order_id repeats because the order has multiple products	OrderItem stores each product line.
amount appears beside payment facts	Payment stores settlement details.

The important point is that modeling does not throw the source data away. It gives the facts a more stable home.

7. The Modeling Process

The process moves from abstract to concrete. Start with the business, define the structure, then generate or write the physical implementation.

- Start conceptually: name the business things and relationships.
- Refine logically: add attributes, keys, cardinality, classifications, and rules.
- Implement physically: create tables, constraints, indexes, security, and loading patterns.
- Review with stakeholders: confirm that the implemented structure still matches the business meaning.

Forward And Reverse Engineering

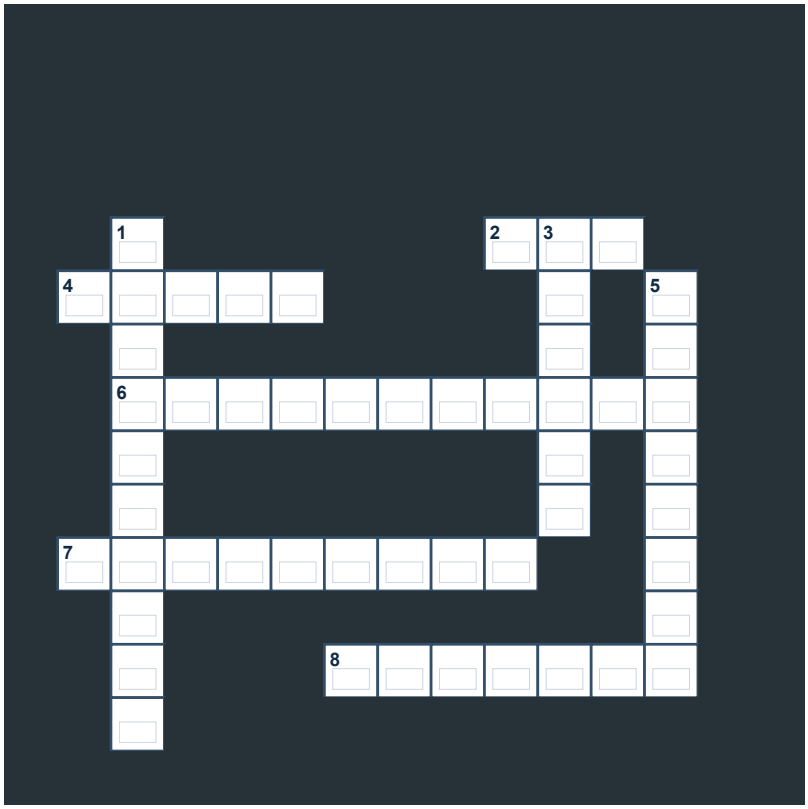
Path	Meaning
Forward engineering	Business understanding becomes conceptual, logical, then physical design.
Reverse engineering	Existing tables, reports, or DDL are studied to recover the conceptual and logical meaning.

In practice, modelers often do both. You may begin with an existing database, infer a logical model, then speak to business users to correct the conceptual meaning.

Quick check: Why might reverse engineering be risky if you do not speak to business users?

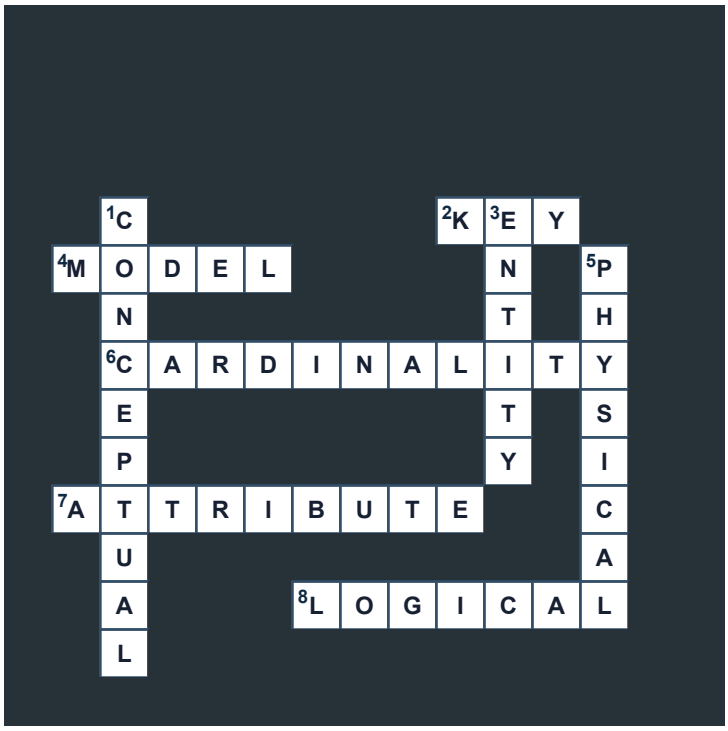
8. One Clean Glossary Crossword

This is the only puzzle in the guide. It is here to reinforce the core vocabulary, not replace the teaching.



No.	Clue
1 Down	Business-level model focused on meaning.
2 Across	Identifier for a row or entity.
3 Down	Business thing worth tracking.
4 Across	Simplified representation of the data world.
5 Down	Implementation model with tables, types, indexes, and constraints.
6 Across	How many of one thing can relate to another.
7 Across	Detail that describes an entity.
8 Across	Technology-independent model with structure and keys.

Answer Key



No.	Answer
1 Down	Conceptual
2 Across	Key
3 Down	Entity
4 Across	Model
5 Down	Physical
6 Across	Cardinality
7 Across	Attribute
8 Across	Logical

Source Notes

Practical Data Modeling article: 'Example - Building a Traditional Data Model'. Used as inspiration for the traditional path from business understanding to conceptual, logical, and physical modeling using a commerce-style example.

Source: https://practicaldatamodeling.substack.com/p/example-building-a-traditional-data

ScaleFree article: 'Conceptual vs Logical vs Physical Data Models' by Michael Olschimke. Used for the three-layer comparison, the sketch/blueprint/build analogy, and the explanation that conceptual, logical, and physical models are steps in one process.

Source: https://www.scalefree.com/knowledge/webinars/data-vault-friday/conceptual-vs-logical-vs-physical-data-models/